

CAPSTONE PROJECT REPORT

SDET TRAINING PROGRAM

PARA BANK AUTOMATION TESTING FRAMEWORK

TEST PLAN : ParaBank Banking Application Automation Testing Project

By : Pankaj Vishwakarma

Github url : https://github.com/Pankajvis1/Capstone-Project-Wipro_SDET

Project Information

Attribute	Details
Project Name	ParaBank Banking Application Automation Framework
Application Under Test (AUT)	ParaBank
Application URL	https://parabank.parasoft.com/parabank
Testing Type	Functional Automation Testing
Automation Tool	Selenium WebDriver
Framework	TestNG
Programming Language	Java
Design Pattern	Page Object Model (POM)
Build Tool	Maven
Reporting Tool	Extent Reports

Attribute	Details
Logging Framework	Log4j2
Version Control	Git & GitHub
CI/CD Tool	Jenkins
Containerization Tool	Docker
Prepared By	Pankaj Vishwakarma

1. Introduction

The banking domain requires highly reliable software systems capable of securely handling customer information and financial transactions. Ensuring the quality, stability, and correctness of such applications is critical.

This Test Plan outlines the strategy, scope, objectives, resources, environment, deliverables, and risk management approach for automating the ParaBank Banking Application using Selenium WebDriver and TestNG.

The automation framework will validate critical banking workflows while ensuring maintainability, scalability, and efficient execution through reporting, logging, CI/CD integration, and containerized execution.

2. Project Objectives

The primary objectives of this automation project are:

Functional Objectives

- Validate customer registration functionality.
- Verify login and logout functionality.
- Validate account management workflows.
- Verify fund transfer operations.
- Validate bill payment processing.
- Verify transaction search functionality.

- Validate customer profile updates.
- Verify loan request processing.

Framework Objectives

- Develop a reusable automation framework.
 - Implement Page Object Model (POM).
 - Support Data Driven Testing.
 - Generate automated execution reports.
 - Implement centralized logging.
 - Enable screenshot capture on failures.
 - Integrate Jenkins CI/CD execution.
 - Support Docker-based execution.
 - Enable cross-browser execution.
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3. Scope of Testing



In Scope

Customer Management

- New Customer Registration
- Registration Validation

Authentication

- Valid Login
- Invalid Login
- Session Validation
- Logout Functionality

Account Services

- Open New Savings Account
- Fund Transfer Between Accounts
- Bill Payment Processing
- Find Transactions
- Update Contact Information
- Request Loan Processing

Framework Features

- Data Driven Testing

- Screenshot Capture
 - Extent Report Generation
 - Log4j2 Logging
 - Jenkins Integration
 - Docker Integration
 - Cross Browser Testing
 - Negative Testing
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Out of Scope

The following activities are not included in the current phase of testing:

- API Testing
 - Database Testing
 - Mobile Application Testing
 - Performance Testing
 - Load Testing
 - Stress Testing
 - Security Testing
 - Penetration Testing
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4. Test Strategy

The project will follow an automation-first approach using Selenium WebDriver and TestNG.

The following testing methodologies will be applied:

Functional Testing

Validation of business workflows against expected application behavior.

Regression Testing

Verification that existing functionality remains unaffected after modifications.

Data Driven Testing

Execution of test cases using external CSV test data.

Cross Browser Testing

Validation of application behavior across supported browsers.

Negative Testing

Validation of application behavior under invalid and unexpected conditions.

5. Test Environment

Component	Configuration
Operating System	Windows / macOS
Browser	Chrome, Safari
Automation Tool	Selenium WebDriver
Framework	TestNG
Programming Language	Java
IDE	Eclipse
Build Tool	Maven
Reporting Tool	Extent Reports
Logging Framework	Log4j2
Version Control	GitHub
CI/CD Tool	Jenkins
Containerization Tool	Docker

6. Test Scenarios

Functional Test Scenarios

Test Case ID	Test Scenario	Priority
TC_REG_001	Register New Customer	High
TC_LOGIN_001	Login with Valid Credentials	High
TC_LOGIN_002	Login with Invalid Username and Password	Medium
TC_LOGIN_003	Login with Invalid Username and Empty Password	Medium
TC_LOGIN_004	Login with Empty Username and Invalid Password	Medium
TC_ACC_001	Open New Savings Account	High
TC_TRF_001	Transfer Funds Between Accounts	High
TC_BILL_001	Bill Payment Processing	High
TC_FIND_001	Find Transactions	High
TC_UPDATE_001	Update Contact Information	High
TC_LOAN_001	Request Loan Processing	High
TC_LOGOUT_001	Logout from Application	High

Planned Negative Test Scenarios

Test Case ID	Scenario	Priority
TC_NEG_001	Transfer Funds with Negative Amount	High
TC_NEG_002	Bill Payment with Negative Amount	High
TC_NEG_003	Transfer Funds to Same Account	High
TC_NEG_004	Request Loan Allows Negative Down Payment	High

7. Test Data Requirements

The following test data will be required during execution.

Registration Data

- First Name
- Last Name
- Address
- City
- State
- Zip Code
- Phone Number
- SSN
- Username
- Password

Login Data

- Valid Credentials
- Invalid Credentials

Banking Data

- Transfer Amount
 - Bill Payment Amount
 - Loan Amount
 - Down Payment Amount
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8. Entry Criteria

Testing activities can begin when the following conditions are met:

- ParaBank application is accessible.
 - Test environment is configured.
 - Selenium framework setup is completed.
 - Required dependencies are available.
 - Test cases are reviewed and approved.
 - Test data is prepared.
 - Jenkins environment is configured.
 - Docker environment is available.
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9. Exit Criteria

Testing activities will be considered complete when:

- All planned test cases are executed.
 - Critical functionalities are validated.
 - Defects are documented.
 - Reports are generated successfully.
 - Test execution summary is prepared.
 - Project deliverables are completed.
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10. Test Deliverables

Documentation Deliverables :

- Test Plan
- Test Cases
- Defect Report
- Test Summary Report
- Final Project Report

Automation Deliverables :

- Selenium Automation Scripts
- Page Object Classes
- TestNG Suite Files
- Data Provider Files

Reporting Deliverables :

- Extent Reports
- TestNG Reports
- Log Files
- Screenshots

DevOps Deliverables :

- Jenkinsfile
- Dockerfile
- Maven Configuration (pom.xml)

11. Risk Analysis and Mitigation

Risk	Impact	Mitigation Strategy
Application Downtime	High	Re-execute tests after recovery
Dynamic Test Data	Medium	Use unique runtime data
Duplicate Username Registration	Medium	Generate dynamic usernames
Browser Compatibility Issues	Medium	Cross-browser validation
Driver Version Mismatch	Medium	Use WebDriverManager
Slow Application Response	Medium	Implement Explicit Waits
Internet Connectivity Issues	Medium	Verify network stability before execution

12. Roles and Responsibilities

Role	Responsibility
Automation Engineer	Framework Design & Development
Tester	Test Case Design & Execution
Developer	Defect Resolution
Project Reviewer	Review and Approval

13. Expected Deliverables

At the completion of the project, the following outputs are expected:

Framework Deliverables

- ✓ Selenium Automation Framework
- ✓ Page Object Model Implementation
- ✓ Data Driven Testing Implementation
- ✓ Reporting and Logging Integration
- ✓ Cross Browser Support

DevOps Deliverables

- ✓ Jenkins CI/CD Integration
- ✓ Docker Containerization Support

Documentation Deliverables

- ✓ Test Plan
 - ✓ Test Cases
 - ✓ Defect Report
 - ✓ Final Project Report
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14. Approval

Name	Role
Pankaj Vishwakarma	SDET Trainee
Vaishali	Trainer/Reviewer

Document Status

 **READY FOR PROJECT EXECUTION**

Prepared By: Pankaj Vishwakarma

Project: ParaBank Banking Application Automation Framework

Document Type: Test Plan